MEMORANDUM

TO Gérard Cachon · Christian Terwiesch
FROM Karl T. Ulrich
DATE 29 July 2026
RE Do early downloads predict eventual citations? — preprint pilot for Paper B (Result 2)

> **Bottom line.** On real per-article data, **early PDF downloads predict a preprint's eventual citations at Spearman ρ ≈ 0.43 (medicine) to 0.47 (biology), and the signal is set within about three months** — squarely in the range of the classic out-of-field benchmarks (Brody, physics ≈ 0.4; Perneger, medicine ≈ 0.5). The effect is real but bounded: early attention rank-orders eventual impact without determining it. That is direct, in-domain support for *admit broadly, decide late*, and it depends on no assumption about AI evaluators. This is a pilot; the economics number awaits a data request to RePEc (below).

Why a preprint pilot

Result 2 asks whether early **download** activity — observable months before any citation can accrue — forecasts eventual citations for working papers. The economics version (a 2015 RePEc cohort, downloads from LogEc) is **blocked on data access**: LogEc's per-item download pages sit under a path its `robots.txt` disallows, so the counts must be **requested** from the RePEc maintainers, not scraped — the same terms-first posture that keeps SSRN off-limits. To validate the method now and anchor the expected effect size, we ran the identical pipeline on the two preprint servers that publish open per-article download data: medRxiv (medicine, the closest analog to Perneger) and bioRxiv (biology).

What we did

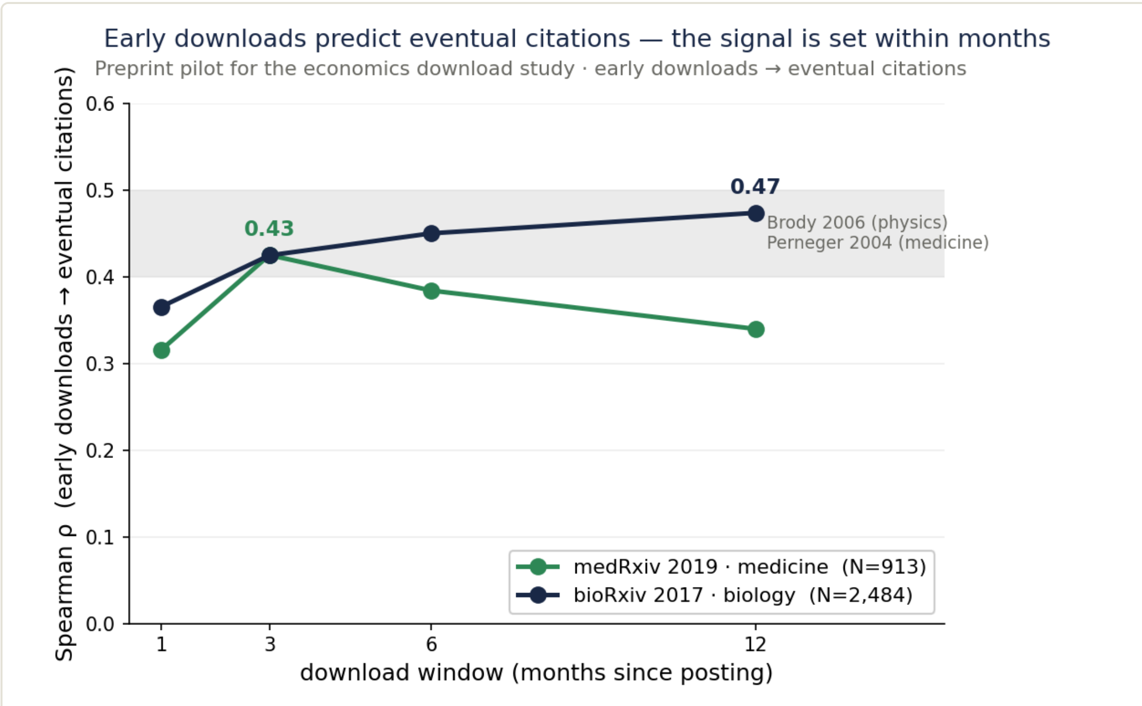
Two matured cohorts: **medRxiv 2019** (medicine, N = 913; pre-COVID, fully matured) and **bioRxiv 2017** (biology, N = 2,484; a larger, longer-matured robustness cohort). Per-article monthly PDF downloads come from the open Rxivist database snapshot (crawl frozen February 2023, so the first-12-month window is fully captured); we cumulate downloads over the first 1, 3, 6, and 12 months from posting. Eventual citations come from OpenAlex, joined by DOI (100% of sampled DOIs matched). For medRxiv, where 79% of the preprints were later published, we use the citations of the **published version of record** — the preprint DOI alone undercounts them roughly tenfold. The statistic is the Spearman correlation between early downloads and eventual citations at each window, plus the tie-robust AUC for identifying the eventual top decile.

Results

COHORT	N	Ρ @1MO	Ρ @3MO	Ρ @6MO	Ρ @12MO	AUC (TOP DECILE)

• **medRxiv 2019 · medicine**	913	0.32	0.43	0.38	0.34	0.79
• **bioRxiv 2017 · biology**	2,484	0.37	0.42	0.45	0.47	0.78

Robustness — the citation measure matters. Because a published preprint's citations mostly accrue to the journal version, the preprint DOI undercounts impact. Using medRxiv's version-of-record citations, median citations rise from 2 to 19 and the correlation rises from ρ = 0.35 (preprint-DOI) to a peak of 0.43. bioRxiv's 0.47 uses the preprint DOI and is therefore a conservative lower bound.



Early downloads → eventual citations, by download window. Shaded band: the Brody (2006) / Perneger (2004) benchmark range. Both cohorts enter it and plateau early.

What it means

- **In the benchmark band, in-domain.** ρ ≈ 0.43–0.47 matches Brody (physics) and Perneger (medicine), now demonstrated on preprint servers with modern per-article data.
- **The signal is set within months.** Both cohorts reach their level by ~3 months and do not sharpen thereafter — an even stronger version of Perneger's plateau.
- **Downloads are a dense signal:** every paper has them (0% zero), unlike citations (24–29% of these papers are still uncited years later), and they appear in the first month — the earliest ex-post signal available.
- **Real but bounded.** The same conclusion as our citation-based early-signal study (ρ ≈ 0.52 at one year on the Management Science cohort), reached from an even earlier vantage point.

Caveats

- **bioRxiv is not version-corrected.** Its ρ = 0.47 uses preprint-DOI citations; building the full bioRxiv-2017 preprint→published map is a ~1-hour API pull, and since the figure is already in-band and the correction only raises it, we left bioRxiv as a stated lower bound. medRxiv *is* version-corrected.
- **medRxiv 2019 is a young, small cohort** (the server launched mid-2019; early adopters), which may still depress its version-corrected ρ of 0.43.
- **Preprints are not working papers.** This validates the method and brackets the effect size; it does not substitute for the economics number.

Next step — the economics result

The pipeline is validated and the citation half of the economics study is already built (we assembled the 1,020-paper NBER 2015 cohort from OpenAlex and verified the download↔citation join key). The one remaining input is the LogEc download data, which must be requested from RePEc. Two decisions: (1) the cohort — a broad set of freely-RePEc-accessible 2015 economics working papers (best download signal) versus NBER 2015 (fast, but LogEc undercounts NBER, which is downloaded mostly direct from nber.org); and (2) whether to send the request now. On your word we finalize the handle list and the email.

Data & reproducibility. Downloads: Rxivist snapshot, Abdill & Blekhman, Zenodo 10.5281/zenodo.7688682 (crawl frozen 2023-02). Citations: OpenAlex (CC0), July 2026. Pipeline, cached cohort, and figure are reproducible in **analysis/download_citation_pilot/**; full writeup in **docs/download_citation_pilot_result.md**. Benchmarks: Brody, Harnad & Carr (2006), *JASIST* 57(8), 1060–1072; Perneger (2004), *BMJ* 329, 546–547.